THE COMPARISON BETWEEN THE ZARISKI TOPOI OF THE BIG AND SMALL SITES OF A LOCALLY QUASI-COMPACT SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the seven items, with two Xy-pic diagrams delivered as written (the square of topoi in the statement and the fibred square in the proof of (7a)).

Editorial note. The label occurs in no other staged Stacks chapter; the checked list is chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. Every labelled reference occurring inside the counted spans points to a chapter of the Project that is not part of this card's source tree. Each is carried in assembly.external_references as a non-load-bearing navigation pointer whose source location is the invocation inside the counted span, so no load-bearing mathematics is replaced by a stub.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The category LC_Zar of locally quasi-compact Hausdorff spaces with the Zariski topology and the morphisms of topoi pi_X. Same chapter, the Zariski topology on LC, the topos of sheaves on LC_Zar/X and the morphism pi_X relating it to the usual topos of sheaves on X. Supplies every object and morphism occurring in the statement.
- Cohomology of the big and small Zariski sites compared by an adjunction. Sites, the results comparing a site with a larger site along a continuous and cocontinuous functor, applied here to the inclusion of the open subsets of X into LC_Zar/X. Supplies the identifications pi_{X,*} = tau_X^{-1} and the identity tau_X^{-1} pi_X^{-1} = id from which parts (1)-(3) are read off.
- Higher direct images on the big Zariski site and their description by a presheaf. Same chapter, the lemma describing the higher direct images R^q f_{Zar,*} by the sheaf associated to a presheaf of cohomology groups on the slices. Supplies the computation in part (7b), where the cohomology sheaves of Rf_{Zar,*}pi_X^{-1}K are identified through that description.

Source reference. Original lemma 03YU: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 646–646. Navigation pointer invoked at proof line

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7812; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00XU: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7818–7818. Navigation pointer invoked at proof line 7818; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 09X3: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7675–7675. Navigation pointer invoked at proof line 7826; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03EH: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7833–7833. Navigation pointer invoked at proof line 7833; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 09W9: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7836–7836. Navigation pointer invoked at proof line 7836; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03CB: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7839–7839. Navigation pointer invoked at proof line 7839; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03F3: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 608–608. Navigation pointer invoked at proof line 7846; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0D90: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7877–7877. Navigation pointer invoked at proof line 7877; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0D6I: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4434–4434. Navigation pointer invoked at proof line 7887; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 01EZ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7907–7907. Navigation pointer invoked at proof line 7907; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 09V6: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7909–7909. Navigation pointer invoked at proof line 7909; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let X be an object of LC_{Zar}. Then*

(1) *for $\mathcal{F} \in Ab(X)$ we have $H^n_{Zar}(X, \pi_X^{-1}\mathcal{F}) = H^n(X, \mathcal{F})$,*
(2) *$\pi_{X,*} : Ab(LC_{Zar}/X) \to Ab(X)$ is exact,*
(3) *the unit $id \to \pi_{X,*} \circ \pi_X^{-1}$ of the adjunction is an isomorphism, and*
(4) *for $K \in D(X)$ the canonical map $K \to R\pi_{X,*}\pi_X^{-1}K$ is an isomorphism.*

Let $f : X \to Y$ be a morphism of LC_{Zar}. Then

(5) *there is a commutative diagram*

$$\begin{array}{ccc} Sh(LC_{Zar}/X) & \xrightarrow{f_{Zar}} & Sh(LC_{Zar}/Y) \\ \downarrow{\scriptstyle \pi_X} & & \downarrow{\scriptstyle \pi_Y} \\ Sh(X_{Zar}) & \xrightarrow{f} & Sh(Y_{Zar}) \end{array}$$

of topoi,

(6) *for $L \in D^+(Y)$ we have $H^n_{Zar}(X, \pi_Y^{-1}L) = H^n(X, f^{-1}L)$,*
(7) *if f is proper, then we have*
   (a) *$\pi_Y^{-1} \circ f_* = f_{Zar,*} \circ \pi_X^{-1}$ as functors $Sh(X) \to Sh(LC_{Zar}/Y)$,*
   (b) *$\pi_Y^{-1} \circ Rf_* = Rf_{Zar,*} \circ \pi_X^{-1}$ as functors $D^+(X) \to D^+(LC_{Zar}/Y)$.*

Proof. Proof of (1). The equality $H^n_{Zar}(X, \pi_X^{-1}\mathcal{F}) = H^n(X, \mathcal{F})$ is a general fact coming from the trivial observation that coverings of X in LC_{Zar} are the same thing as open coverings of X. The reader who wishes to see a detailed proof should apply Lemma 03YU to the functor $X_{Zar} \to LC_{Zar}$.

Proof of (2). This is true because $\pi_{X,*} = \tau_X^{-1}$ for some morphism of topoi $\tau_X : Sh(X_{Zar}) \to Sh(LC_{Zar})$ as follows from Sites, Lemma 00XU applied to the functor $X_{Zar} \to LC_{Zar}/X$ used to define π_X.

Proof of (3). This is true because $\tau_X^{-1} \circ \pi_X^{-1}$ is the identity functor by Sites, Lemma 00XU. Or you can deduce it from the explicit description of π_X^{-1} in Lemma 09X3.

Proof of (4). Apply (3) to an complex of abelian sheaves representing K.

Proof of (5). The morphism of topoi f_{Zar} comes from an application of Sites, Lemma 03EH and in our case comes from the continuous functor $Z/Y \mapsto Z \times_Y X/X$ by Sites, Lemma 09W9. The diagram commutes simply because the corresponding continuous functors compose correctly (see Sites, Lemma 03CB).

Proof of (6). We have $H^n_{Zar}(X, \pi_Y^{-1}\mathcal{G}) = H^n_{Zar}(X, f_{Zar}^{-1}\pi_Y^{-1}\mathcal{G})$ for $\mathcal{G}$ in $Ab(Y)$, see Lemma 03F3. This is equal to $H^n_{Zar}(X, \pi_X^{-1}f^{-1}\mathcal{G})$ by the commutativity of the diagram in (5). Hence we conclude by (1) in the case L consists of a single sheaf in degree 0. The general case follows by representing L by a bounded below complex of abelian sheaves.

Proof of (7a). Let $\mathcal{F}$ be a sheaf on X. Let $g : Z \to Y$ be an object of LC_{Zar}/Y. Consider the fibre product

$$\begin{array}{ccc} Z' & \xrightarrow{f'} & Z \\ \downarrow{\scriptstyle g'} & & \downarrow{\scriptstyle g} \\ X & \xrightarrow{f} & Y \end{array}$$

Then we have

$$(f_{Zar,*}\pi_X^{-1}\mathcal{F})(Z/Y) = (\pi_X^{-1}\mathcal{F})(Z'/X) = \Gamma(Z', (g')^{-1}\mathcal{F}) = \Gamma(Z, f'_*(g')^{-1}\mathcal{F})$$

the second equality by Lemma 09X3. On the other hand

$$(\pi_Y^{-1}f_*\mathcal{F})(Z/Y) = \Gamma(Z, g^{-1}f_*\mathcal{F})$$

again by Lemma 09X3. Hence by proper base change for sheaves of sets (Cohomology, Lemma 0D90) we conclude the two sets are canonically isomorphic. The isomorphism is compatible with restriction mappings and defines an isomorphism $\pi_Y^{-1}f_*\mathcal{F} = f_{Zar,*}\pi_X^{-1}\mathcal{F}$. Thus an isomorphism of functors $\pi_Y^{-1} \circ f_* = f_{Zar,*} \circ \pi_X^{-1}$.

Proof of (7b). Let $K \in D^+(X)$. By Lemma 0D6I the nth cohomology sheaf of $Rf_{Zar,*}\pi_X^{-1}K$ is the sheaf associated to the presheaf

$$(g : Z \to Y) \longmapsto H^n_{Zar}(Z', \pi_X^{-1}K)$$

with notation as above. Observe that

$$\begin{aligned} H^n_{Zar}(Z', \pi_X^{-1}K) &= H^n(Z', (g')^{-1}K) \\ &= H^n(Z, Rf'_*(g')^{-1}K) \\ &= H^n(Z, g^{-1}Rf_*K) \\ &= H^n_{Zar}(Z, \pi_Y^{-1}Rf_*K) \end{aligned}$$

The first equality is (6) applied to K and $g' : Z' \to X$. The second equality is Leray for $f' : Z' \to Z$ (Cohomology, Lemma 01EZ). The third equality is the proper base change theorem (Cohomology, Theorem 09V6). The fourth equality is (6) applied to $g : Z \to Y$ and Rf_*K. Thus $Rf_{Zar,*}\pi_X^{-1}K$ and $\pi_Y^{-1}Rf_*K$ have the same cohomology sheaves. We omit the verification that the canonical base change map $\pi_Y^{-1}Rf_*K \to Rf_{Zar,*}\pi_X^{-1}K$ induces this isomorphism. □